

EDS White Paper: The Unforeseen Benefits of Sewer Inflow & Infiltration Monitoring

Environmental Data Services

Where Quality Flows...

Executive Summary

Sewer inflow and infiltration (I/I) monitoring has long been an essential tool for utilities to assess stormwater ingress, calibrate hydraulic models and prioritise rehabilitation works. However, recent projects across Australia have revealed that these programs often deliver a far wider range of operational, environmental and financial benefits than originally intended.

From identifying network blockages and verifying maintenance effectiveness to detecting illegal discharges and optimising pumping operations, the unexpected insights gained from accurate flow monitoring are transforming how utilities understand and manage their networks. This paper explores these unforeseen advantages and demonstrates how utilities are leveraging monitoring data to achieve outcomes that extend well beyond traditional inflow and infiltration objectives.

1. Introduction

Utilities invest heavily in I/I monitoring programs to better understand rainfall dependent inflow, infiltration and system capacity. Traditionally, these studies aim to quantify stormwater ingress, calibrate hydraulic models and guide targeted capital improvements.

With the introduction of high precision monitoring technologies such as the [Detectronic MSFM Area Velocity Flow Meter](#) and [LIDoTT Level Sensor](#), combined with advanced cloud based data platforms like **DetectData**, utilities now have access to a level of detail that was previously unattainable.

This higher data fidelity has led to the discovery of secondary (and often more immediately valuable) benefits. These insights are reshaping asset management, operational decision making and cross departmental collaboration for Australia's water utilities.

2. Unforeseen Benefits Identified by Utilities

2.1 Asset Condition Insights

Accurate depth and velocity data have allowed utilities to detect network blockages, sediment build up, and structural defects long before field crews report them. In one regional council, a sudden shift in the depth-velocity relationship revealed a partial obstruction in a rising main, prompting a targeted CCTV inspection that confirmed debris accumulation.

Similarly, depth anomalies have highlighted sags, displaced joints and infiltration points, enabling proactive maintenance rather than reactive response.

2.2 Verification of Maintenance & Contractor Works

Flow monitoring data provides measurable proof of maintenance effectiveness. Following a cleaning and relining program in a coastal network, flow profiles recorded by EDS showed a return to baseline hydraulic behaviour within 24 hours, verifying that the works restored full capacity.

Utilities are increasingly using these before and after comparisons to verify contractor performance, reducing disputes and improving accountability across maintenance programs.

2.3 Detection of Cross Connections & Illegal Discharges

Dry weather flow signatures have helped utilities detect unexpected industrial inflows and cross connections. In one Queensland catchment, persistent overnight flow despite zero rainfall led to the discovery of a trade waste connection incorrectly tied into the sewer network.

Similarly, sustained flow increases following rainfall events in non-residential areas often indicate inflow from adjacent stormwater systems or unapproved private connections. Identifying and addressing these issues not only reduces treatment costs but also improves compliance and system resilience.

2.4 Operational Optimisation

Continuous monitoring has allowed utilities to optimise pump station operations by correlating real time flow data with rainfall and catchment response. In Sydney's outer suburbs, operators utilised flow data from LIDoTT Smart sensors to refine pump start/stop levels, reducing both run time and power consumption.

This level of operational insight, once unavailable through SCADA alone, enables smarter, data driven decisions that improve network efficiency and extend asset life.

2.5 Targeted Investment & Planning

With reliable flow data, utilities can now prioritise capital works based on actual network performance rather than assumptions. By mapping inflow severity across catchments, authorities are able to direct CCTV inspections and rehabilitation budgets precisely where they're needed.

This data led approach was successfully applied by a regional NSW council, resulting in a 30% reduction in unnecessary relining expenditure and faster achievement of compliance targets.

2.6 Environmental & Community Outcomes

Reducing wet weather inflow directly benefits both the environment and the community. Lower treatment volumes mean decreased energy consumption and greenhouse gas emissions. Fewer overflow events translate to improved waterway health and reduced community disruption.

These broader environmental benefits often form part of utilities' sustainability reporting, demonstrating the value of flow monitoring beyond compliance, contributing directly to net zero and climate resilience objectives.

2.7 Organisational Collaboration & Data Integration

Perhaps the most overlooked outcomes of modern I/I programs is the way data promotes collaboration across departments. Flow data visualised through platforms such as DetectData enables engineers, field technicians, asset planners and environmental officers to work from the same dataset, aligning objectives and accelerating decision making.

This cross functional integration ensures that lessons from one project inform the next, creating a continuous improvement cycle within the utility.

3. Technology Enablers

The emergence of highly accurate, intrinsically safe flow meters such as the [Detectronic MSFM AV](#) and the [LIDoTT Smart](#) & [Alarm range](#) has made it possible to collect dependable, long term data even in harsh sewer environments.

These instruments, supported by cloud based analytics and API connectivity, are enabling real time data access, rapid fault detection and automated alerts. When paired with EDS's highly trained field crews, utilities can achieve near perfect data integrity, ensuring that every insight gained is based on reliable, validated measurement.

4. Case Study: Regional Council I&I Reduction

Case Study 1 – Verifying Rehabilitation Effectiveness

A major metropolitan utility conducted post maintenance flow monitoring to confirm the success of a relining program. Depth and flow variation during rainfall decreased by more than 40%, validating that the rehabilitation successfully reduced infiltration and confirming modelled performance improvements.

Case Study 2 – Identifying Stormwater Cross Connections

In a large urban catchment, flow data displayed rainfall correlation patterns inconsistent with residential inflow expectations. Subsequent field investigation confirmed that several stormwater lines had been inadvertently connected to the sewer network. Prompt rectification led to a marked reduction in wet weather peak flow volumes.

Case Study 3 – Sediment Detection & Preventative Maintenance

Flow monitoring over several weeks revealed a steady rise in base depth levels without rainfall influence. Targeted inspection found sediment build up at the downstream junction, allowing maintenance teams to clear the line before any surcharge or overflow occurred during the following storm event.

Case Study 4 – Model Calibration & Catchment Reassessment

A regional water authority used temporary flow monitoring to recalibrate its hydraulic model. Data revealed that actual infiltration rates were significantly lower than initially modelled, enabling reallocation of rehabilitation funds and a more accurate prioritisation of high risk catchments.

5. Recommendations

Adopt Continuous Monitoring: Treat flow data as an ongoing operational tool rather than a temporary project.

Integrate Data Across Teams: Encourage collaboration between planning, maintenance and environmental units.

Verify Maintenance Outcomes: Use flow data to validate and document the success of field works.

Focus on Training and QA: Ensure technicians are trained in installation and calibration to maintain data quality.

Leverage Cloud Analytics: Use platforms capable of detecting trends, anomalies, and correlations automatically.

Conclusion

The true value of sewer inflow and infiltration monitoring extends far beyond identifying stormwater ingress. Accurate, continuous flow data is now an operational asset driving smarter maintenance, improving environmental performance, and enhancing the accountability of every network intervention.

Utilities that recognise and act on these unforeseen benefits are not only reducing costs and improving efficiency but also building resilient, data driven organisations capable of meeting the complex challenges of modern water management.

About EDS

Environmental Data Services (EDS) is Australia's trusted partner in sewer flow monitoring and data driven network management. Our experienced field team, cutting edge equipment and ISO compliant data validation processes deliver the reliable insights needed for smarter infrastructure decisions.

Environmental Data Services

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